

1. Explain the different memory areas managed by JVM.

① Stack Memory :

- Stack memory in Java used for static memory allocation & the execution of thread.
- It contains primitive types (values) that are specific to method and references to objects referred from the method that are in heap.
- Access this memory in LIFO order.

② Heap Memory :

- Heap space is used for dynamically memory allocation of Java objects & JRE classes at runtime.
- New objects are always created in heap.
- But these references are stored in heap.

③ Method Area :

- It's a logical part of heap when it is created JVM starts.
- Method Area is used to store class level info such as class structures, static variables, Const pool, interface.

Program Counter :

- Each JVM thread has a program counter.
- it stores address of current JVM instruction.
- For native methods pc value is undefined.

Native method stack :

- native method stacks are not written in Java code.
- This memory is allocated for each thread when it is created & can have either fixed or dynamic size.

② Difference Between Stack & Heap Memory with examples.

<u>sol:-</u> <u>Heap :</u>	<u>stack</u>
→ Stores obj's and instance variables	→ stores method calls & local variables.
Lies	→ LIFO
→ Random Access	→ short lived.
→ Long lived	→ each thread has its own stack.
→ shared among all threads	→ Allocation and deallocation both are automatic
→ Allocation is manual & deallocation is automatic	

③ What is Garbage collection? How does it work in Java?

⇒ Garbage collection in Java is an automatic memory management process that helps Java programs run efficiently.

→ Objects are created on the heap area.

→ Eventually some objects are no longer needed.

→ Garbage collection is an automatic process that removes unused objects from heap.

Working:

→ it identifies which objects are still in use and which are not used.

→ it removes the objects that are unreachable.

→ The programmer does not need to mark objects to be deleted explicitly. Garbage collection is implemented in JVM.

④ What are the different types of Garbage collectors available in Java? mention any two?

⇒ Types of Garbage Collectors

1. Serial Garbage Collector
2. Parallel Garbage Collector.

1. Serial Garbage collector: it is the simplest type of Garbage collector.

→ uses single thread to handle all collection tasks.

→ When garbage collection is triggered, it pauses all running threads.

→ it is used for smaller Apps.

2. Parallel Garbage Collector:

→ it is default Garbage Collector in Java8.

→ it uses multiple threads to perform Garbage collection tasks.

→ it is suitable for Apps that need high performance can tolerate some pauses.

→ it is better than the Serial Garbage Collector.

⑤ What is the role of finalize () method in Java? Is it recommended to use it? Why/Why not.

Sol: The finalize () Method in Java is called by the Garbage collector just before an object is destroyed.

→ it allows the obj to perform cleanup Activity. Cleanup Activity means closing the resource associated with it object. like database connection.

→ The finalize () method in Java 9 has been deprecated & Java 18 it is removed.

→ it is not recommended in modern Java applications. prefer alternatives like try-with resources for resource management and cleanup.